WEEK 1: REVISION & WEBSITE PLANNING

Revision

Students already know:

- Semantic HTML
- External CSS
- Flexbox
- Grid
- Page layouts

Website Planning

Before building a website, we must plan:

- What is the purpose?
- Who will use it?
- What pages are needed?

Wireframe

A **wireframe** is a simple drawing of how a webpage will look.

Example pages:

- Home
- About
- Contact
- Dashboard

Folder Structure

project/

├── index.html

├── about.html

├── css/

│ └── style.css

└── js/

└── script.js

Class Activity

- Draw a wireframe on paper
- Create project folders

WEEK 2: RESPONSIVE WEB DESIGN (INTRODUCTION)

What is Responsive Design?

Responsive design means a website **looks good on phones, tablets, and computers**.

Viewport Meta Tag

<meta name="viewport" content="width=device-width, initial-scale=1.0">

Flexible Units

- %
- vw (viewport width)
- vh (viewport height)

Example:

```
.container {
  width: 90%;
}
```

Class Activity

Resize browser window and observe layout changes.

WEEK 3: MEDIA QUERIES

What is a Media Query?

Media queries allow CSS to change based on screen size.

Basic Media Query

```
@media (max-width: 600px) {
  body {
    background-color: lightgray;
  }
}
```

Breakpoints

- Mobile: up to 600px
- Tablet: 601px – 900px
- Desktop: above 900px

Class Activity

Change font size for mobile screens.

WEEK 4: ADVANCED CSS SELECTORS

Descendant Selector

```
div p {
  color: blue;
}
```

Child Selector

```
div > p {
  color: red;
}
```

Attribute Selector

```
input[type="text"] {
  border: 2px solid green;
}
```

Pseudo-Classes

```
a:hover {
  color: red;
}
```

Class Activity

Add hover effects to links and buttons.

WEEK 5: CSS TRANSITIONS & ANIMATIONS

Transitions

Transitions make changes smooth.

```
button {
  background-color: blue;
  transition: background-color 0.5s;
}
```

```
button:hover {
  background-color: green;
}
```

Transforms

```
div:hover {
  transform: scale(1.1);
}
```

Keyframe Animation

```
@keyframes move {
  from { left: 0; }
  to { left: 100px; }
}
```

Class Activity

Create a button with hover animation.

WEEK 6: FORMS – ADVANCED STYLING & LAYOUT

Multi-Section Forms

Use <fieldset> and <legend>.

Form Layout with Flexbox

```
form {
  display: flex;
  flex-direction: column;
}
```

HTML Validation

<input type="email" required>

Class Activity

Design a **student registration form**.

WEEK 7: INTRODUCTION TO JAVASCRIPT

What is JavaScript?

JavaScript makes websites **interactive**.

Uses of JavaScript

- Show messages
- Respond to clicks
- Validate forms

Linking JavaScript

<script src="js/script.js"></script>

First JavaScript Code

alert("Welcome to my website");

Class Activity

Show an alert when page loads.

WEEK 8: JAVASCRIPT VARIABLES & EVENTS

Variables

```
let name = "John";

const age = 15;
```

Click Event

```
<button onclick="showMessage()">Click Me</button>

function showMessage() {
  alert("Button clicked!");
}
```

Class Activity

Create a button that displays a message.

WEEK 9: JAVASCRIPT CONDITIONS

If Statement

```
let age = 18;


if (age >= 18) {
  alert("You are eligible");
} else {
  alert("Not eligible");
}
```

Simple Login Logic

```
if (username === "admin") {
  alert("Login successful");
}
```

Class Activity

Create a simple login check using if/else.

WEEK 10: DOM MANIPULATION (BASICS)

What is DOM?

DOM means **Document Object Model**.
It allows JavaScript to change HTML content.

Accessing Elements

```
document.getElementById("text").innerHTML = "Hello";
```

Changing Styles

```
document.getElementById("box").style.backgroundColor = "red";
```

Show & Hide

```
element.style.display = "none";
```

Class Activity

Create a button that hides text.

WEEK 11: MINI PROJECT – INTERACTIVE WEBSITE

Project Example

Student Dashboard Website

Features:

- Login page
- Dashboard page
- Styled layout

- Button interactions
- Responsive design

Tools Used

- HTML
- External CSS
- JavaScript

WEEK 12: PROJECT COMPLETION & DEBUGGING

Debugging

- Fix layout issues
- Correct JavaScript errors
- Test responsiveness

Code Improvement

- Add comments
- Rename variables properly
- Remove unused code